Software Agile Project Management Composition

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For more information, please refer to the original source:
Book Title: Agile Project Manager_v2_handbook
Author: Agile Business Consortium
Publisher: Buckland Media Group Limited
Year of Publication: 2017
Please consult the book for further details and complete content.

Outline

- Introduction to Dynamic Systems Development Method (DSDM)
- The composition of DSDM
- DSDM and Scrum
- Comparison among Scrum, Kanban, and Scrumban
- Introduction to Project Management Tools

Introduction : Dynamic Systems Development Method (DSDM)

Recap…

The DSDM philosophy is that:

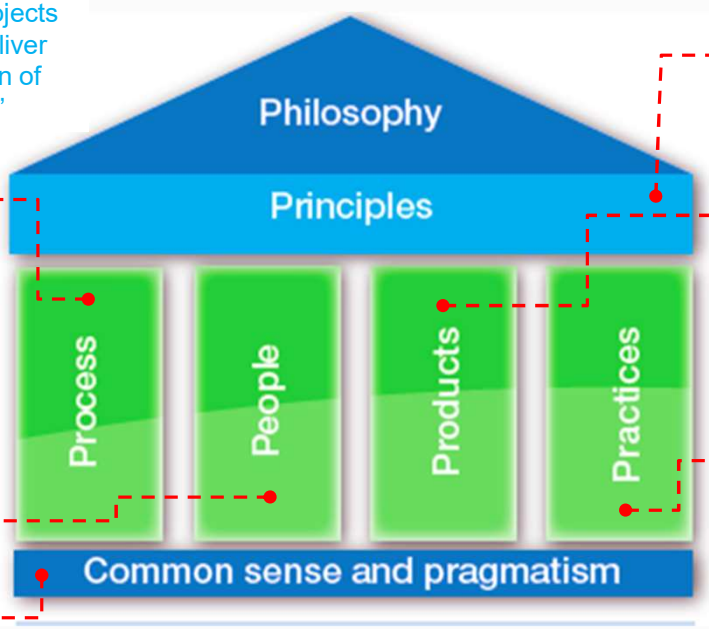
"best business value emerges when projects are aligned to clear business goals, deliver frequently and involve the collaboration of motivated and empowered people."

This is achieved when

- **Clear Roles & Responsibilities**: DSDM assigns well-defined roles covering business, technical, management, and process interests to ensure balanced representation across the project.
- **Collaboration & Communication**: All team members work closely together, breaking down barriers and fostering effective communication to drive project success.
- **Empowered, Self-Organising Teams**: The best solutions come from teams that are empowered to self-organise and take responsibility within agreed boundaries.

The Composition of DSDM

"best business value emerges when projects are aligned to clear business goals, deliver frequently and involve the collaboration of motivated and empowered people."



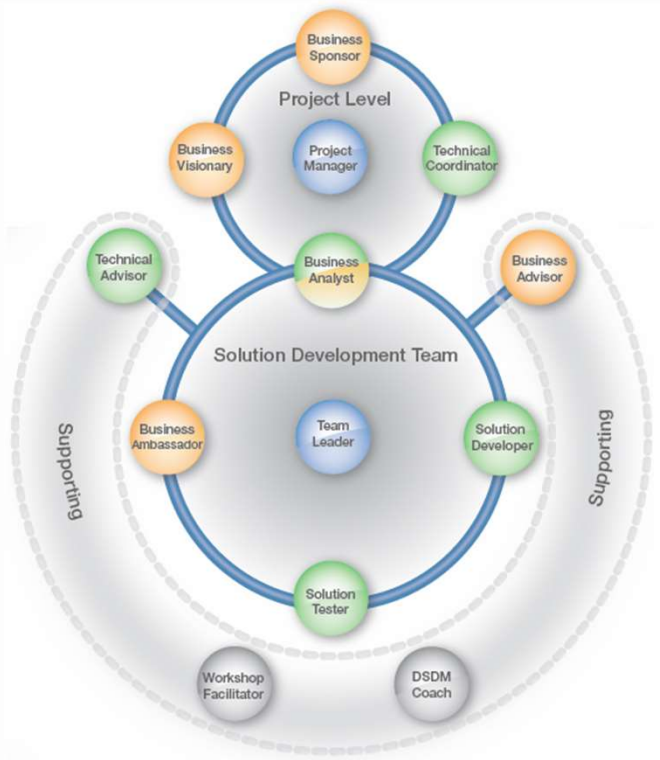
- A set of 8 principles that bring the philosophy into practice.
- The lifecycle, phases, timeboxing, iteration, governance flow. Describes how work is structured.
- The deliverables, artefacts, outputs (business, solution, governance) required. Defines what is produced and tracked.
- Roles, responsibilities, team interactions, empowered decision-making. Captures who does what, and how people collaborate.
- Recommended techniques, methods, workshops, modelling, testing, etc. These are tools & approaches to support execution.
- **Common sense**: using sound practical judgment, independent of over-theory.
- **Pragmatism**: making decisions based on immediate practical consequences rather than rigid theory or dogma.

The People in DSDM

- **Colour Scheme (Roles of Interest):** Orange = Business view, Green = Solution/technical view, Blue = Management/leadership view, **Grey** = Process view, Mixed = Covers multiple views.

- **Role Categories:**
  - **Project-level roles**: Business Sponsor, Business Visionary, Technical Coordinator, Project Manager, Business Analyst → direct, manage, and govern the project.
  - **Solution Development Team roles**: Business Ambassador, Solution Developer, Solution Tester, Business Analyst, Team Leader → engine room of the project, shaping and building the solution.
  - **Supporting roles**: Business Advisors, Technical Advisors, Workshop Facilitator, DSDM Coach → provide guidance, expertise, and support as needed.

- **Role Flexibility:** One person can hold multiple roles, or a role may be shared by several people, provided communication and collaboration remain effective.



DSDM and Scrum

Aspect	DSDM	Scrum
Scope / Lifecycle Coverage	Covers the entire project lifecycle: pre-project, feasibility, foundations, evolutionary development, deployment, post-project.	Primarily focused on the development lifecycle (product backlog → sprints → increments). Governance / project initiation phases are less prescribed.
Governance & Structure	More structured governance, roles, disciplines, and artefacts. Designed to balance agility with control.	Lightweight framework with minimal prescribed governance outside the core roles and events.
Fixed vs Variable Elements	Fixes time, cost, quality early on; scope is variable and managed via MoSCoW prioritisation.	Fixes scope (commitment) per sprint; time is fixed in each sprint, cost is implicit (team capacity).
Requirements / Scope Management	Uses MoSCoW prioritisation to categorise requirements (Must, Should, Could, Won't).	Uses product backlog ordering / prioritisation; changes may be made between sprints but generally backlog is fluid.
Roles & Role Variety	Many roles: Business Sponsor, Business Visionary, Technical Coordinator, Business Analyst, Solution Developer, etc. Rich role structure.	Fewer roles: Product Owner, Scrum Master, Development Team. More lean and focused.
Change Handling	Accepts change, but within boundaries; scope is trimmed if needed (using MoSCoW) to meet time and cost constraints.	Encourages change, backlog is adaptive; within a sprint, changes are limited to preserve focus.
Best Use / Fit	Larger projects needing agility plus governance — especially where project initiation, accountability, stakeholder roles, and full lifecycle governance are needed.	Product development teams wanting agility, fast feedback, minimal overhead, and frequent delivery in iterative cycles.
Complexity & Overhead	Higher overhead because of more roles, more governance, more artefacts and control points.	Lower overhead; lean structure with less prescribed artefacts and roles.

Comparison among Scrum, Kanban, and Scrumban

Aspect	Scrum	Kanban	Scrumban
Description	Lightweight framework to develop and sustain complex products via defined accountabilities, events, and artifacts.	Visual flow method that optimizes work by visualizing tasks and limiting WIP to keep work moving.	Hybrid that combines Scrum's cadence/ceremonies with Kanban's visual flow and WIP limits for more flexibility.
Iterations/Time boxing	Work happens in fixed-length Sprints of one month or less. (Many teams use 1–4 weeks.)	No fixed iterations; teams pull work continuously.	Uses short sprints when helpful plus continuous pull as capacity allows.
Roles & Events	Explicit events: Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective (within each Sprint).	Few mandated roles/ceremonies; teams keep it lightweight and adapt meetings to need.	May keep some Scrum ceremonies (e.g., standups/retros) while applying Kanban flow controls.
Work in Progress (WIP)	WIP is indirectly constrained by the Sprint commitment and Sprint Goal.	Explicit WIP limits per workflow stage to expose bottlenecks and stabilize flow.	Uses Kanban-style WIP limits with simple policies/triggers to manage flow.
Planning & Adaptation	Plan at the start of each Sprint; avoid changes that endanger the Sprint Goal during the Sprint.	Continuous prioritization; new work is pulled as capacity appears.	On-demand planning triggered when the ready backlog drops below a threshold.
Best Use Case	Cross-functional product teams needing structure and a steady delivery cadence.	Ops/support/maintenance or any flow-based work needing adaptability and smooth throughput.	Teams that want a balance of structure and flexibility, or are evolving from Scrum↔Kanban.

Introduction to Project Management Tools

Project Management Tools

- Project management tools are software or systems that support planning, executing, controlling, and closing of projects.
- It help teams organize tasks, monitor progress, allocate resources, communicate, and report status.
- These tools act as enablers — they don't replace good process or governance, but make applying them easier, more visible and efficient.
- The choice of tool depends on factors like project complexity, team size, methodology (Agile, Waterfall, hybrid), and integration needs.
- Here are common categories, with their focus and key features:

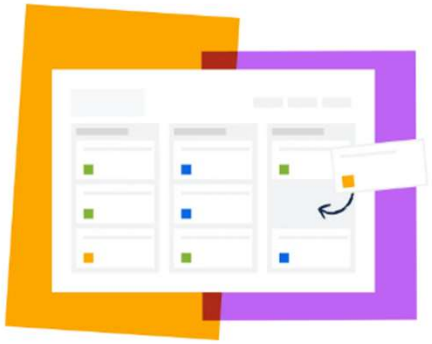
Category	Focus / Purpose	Key Features
All-in-One / Integrated Tools	Provide end-to-end project management coverage	Task planning , scheduling, resource allocation, dashboards, reporting, collaborating.
Task / To-Do Management Tools	Managing individual tasks and simple workflows	Task lists, assignment, due dates, status tracking
Time / Schedule Tools	Focus on timelines and time tracking	Gantt charts, calendars, despendencies, milestones
Resource Management Tools	Allocating and balancing resources	Workload view, availability, conflicts, forecasts
Collaboration / Communication Tools	Enabling team communication and document sharing	Messaging, file sharing, comment threads, versioning
Speacilized / Industry – Specific Tools	Tailored team specific domains or methodologies	For software: backlog + issue tracking for construction: equipment and materials planning: for marketing : campaign timelines
On-Premise vc Cloud / Deployment Types	Based on how the tool is hosted and accessed	Installed servers (on-premise) vs Saas / cloud tools; mobile or desktop clients

Comparison of the Project Management Tools

Tool	Strengths / Key Features	Weaknesses / Trade-offs	Best Fit / Use Case
Jira	Strong support for Agile (Scrum, Kanban ,hybrid) with boards, backlogs, roadmaps, and integrations. Powerful automation templates and workflow customization. Excellent issue / fast tracking, reporting, and visibility across teams.	Can be complex to configure and maintain for non-technical teams. Some features (reporting, advanced planning) may require add-ons or plugins.	Software development teams. Agile teams needing robust issue tracking and process control.
Trello	Simplicity and ease of use ; intuitive Kanban board interface ideal for small teams. Lightweight fast setup, minimal training required	Limited built-in advanced features (reporting, resource management)- relies heavily on integrations of 'Power-Ups'. Less suitable for large or complex projects without customization or add-ons.	Small to medium teams. Simple or visual projects, task tracking.
Clickup	Rich feature set : docs, goals, time tracking , customize workflows, dashboards. Highly customizable with many view options (list board, Gantt, etc.).	Steeper learning curve and complexity in setup. Overhead to maintain configurations and avoid feature overload.	Teams that want an" all-in –one" platform replacing many tools, scaling beyond simple task tracking.

Jira Tool

Project management



Kanban

Manage and track agile work plus integrate developer tools like GitHub



Project management

Designed to help business teams manage work with list and calendar views



Scrum

Plan, prioritize, and schedule sprints using scrum framework

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Summary

Philosophy: The guiding belief — “best business value emerges when projects align with clear business goals, deliver frequently, and involve empowered people.”

Principles: Eight principles serve as the bridge between philosophy and practice, enforcing the mindset and behaviors required to make DSDM effective.

Four Supporting Domains: The philosophy + principles are supported by:

- Process: Lifecycle, phases, timeboxing, governance flows — how work is structured
- People: Roles, responsibilities, collaboration, decision making — who does what
- Products: Deliverables, artefacts, outputs across business, solution, governance — what is produced
- Practices: Techniques, workshops, modelling, testing, other methods — how work is done

Assessment – WBL Project

Assessment	Example of Case study	Instruction	Rubric
Assignment 1 – User Stories	The university wants to build an e-learning platform for online courses, quizzes, and course materials. Requires functional & non-functional Specification	▪ Write a set of 8–12 user stories (As a <role> I want <feature> so that <benefit>) for key functions (course enrolment, content upload, assessment).	▪ Clarity & Format (user stories well formed) ▪ Coverage (important system aspects captured) ▪ Logical Cohesiveness (stories fit together)
Assignment 2 – MoSCow Classification	From the same e-learning case, refine the user stories and classify them properly.	▪ Prioritise US using MoSCoW (Must, Should, Could, Won't). ▪ Justify each classification with business value / technical risk. ▪ Propose a Minimum Usable Subset (MUS).	▪ Correct Prioritisation (justified MoSCoW classification) ▪ Balance in MUS (reasonable Must vs Should / Could) Alignment with Case Business Goals Consistency & Logic
Assignment 3 – Time-Boxing	Use the previously prioritised stories for time-boxing.	▪ Partition user stories into 2–3 timeboxes (iterations) of fixed duration (e.g. 2 weeks each). ▪ Assign stories to timeboxes, ensure balance. ▪ Provide a simple schedule / plan showing which features delivered when.	▪ Feasibility (workload per timebox realistic) ▪ Cohesion (dependencies respected) Value Delivered Early (Musts in first boxes) ▪ Clarity of Plan / Schedule
Assignment 4 – Monetization Planning	For the e-learning system, design how the platform can generate revenue or financial sustainability.	▪ Propose 2–3 monetization strategies (e.g. subscription, freemium, course fees, ads). ▪ Model expected income, costs, breaking-even point. ▪ Discuss risks and how Agile delivery interacts with monetization (e.g. feature gating, MVP).	▪ **Business Viability** (realistic financial model) ▪ **Creativity / Innovation** of revenue ideas ▪ **Risk & Mitigation** ▪ **Integration with Agile / delivery plan**

Assignment 1 – User Stories

🎯 Example User Stories (E-learning System)

1. **As a student**, I want to **browse available courses by category or keyword** so that I can find courses relevant to my interests.
   Acceptance Criteria:
   - Course list shows title, short description, instructor name, thumbnail.
   - Search by keyword and filter by subject area works.
   - Clicking a course shows detailed course info page.
2. **As a student**, I want to **enroll in a course** so that I gain access to the course content and assignments.
   Acceptance Criteria:
   - Enrollment button enabled for registered students only.
   - Upon enrollment, "My Courses" list updates immediately.
   - Confirmation email sent to student.
3. **As an instructor**, I want to **upload course materials (videos, PDFs, slide decks)** so that students have learning resources.
   Acceptance Criteria:
   - Instructor sees an "Upload Material" interface.
   - Supported file types: PDF, MP4, PPTX.
   - Uploaded file appears under course module and is accessible by enrolled students.
4. **As an instructor**, I want to **create quizzes or assessments** so that I can evaluate students' understanding.
   Acceptance Criteria:
   - Quiz builder supports multiple choice, short answer, true/false.
   - Quiz can be assigned to specific modules.
   - Students can submit quiz and receive immediate score feedback, or instructor grading.
5. **As a student**, I want to **submit assignments online** so that I don't need to turn in physical copies.
   Acceptance Criteria:
   - Students can upload their answers/documents via assignment interface.
   - System prevents submission after the deadline.
   - Instructor sees submissions list and can grade them.
6. **As an administrator**, I want **dashboard analytics and usage reports** so that I can monitor system engagement and performance.
   Acceptance Criteria:
   - Dashboard shows number of active users, course enrollments, completion rates.
   - Reports can be filtered by date range, course, instructor.
   - Admin can export to CSV.
7. **As a student**, I want **email or in-app notifications** when new announcements or deadlines are posted so that I stay updated.
   Acceptance Criteria:
   - Notification settings allow ON/OFF for email and in-app.
   - When instructor publishes announcement, enrolled students receive notification.
   - Student notification center shows unread alerts.
8. **As a student**, I want the system to **load pages within 2 seconds** so that the user experience feels responsive. (*Non-functional*)
   Acceptance Criteria:
   - 95% of page loads occur under 2 seconds (under typical load).
   - Tool will monitor, log and flag slower pages.
9. **As a guest visitor**, I want to **preview a sample lesson** so that I can evaluate whether to enroll.
   Acceptance Criteria:
   - Guest can view one module sample (e.g. first video or slide).
   - No personal data required to access preview.
   - Preview shows "Enroll to continue" prompt.
10. **As an instructor**, I want to **issue a certificate upon course completion** so that students gain credential recognition.
    Acceptance Criteria:
    - Certificate generation module exists.
    - Students who meet clearance (score threshold, completion) get certificate.
    - Certificate is downloadable PDF, with student name, course name, date.

Assignment 2 - MoSCow Classification

User Story	MoSCoW Category	Justification / Notes
As a student I want to browse available courses so that I can decide which to enroll in.	Must	Core functionality needed for users to see what's offered; without it the system is unusable.
As a student I want to enroll in a course so that I get access to content and assignments.	Must	Central to the platform—no value if students cannot enroll.
As an instructor I want to upload course materials (videos, PDFs) so that students can access learning resources.	Must	Essential for content delivery.
As an instructor I want to create quizzes/assessments so that I can evaluate students' understanding.	Must	Assessment capability is key for learning outcomes.
As a student I want to submit assignments online so I don't need physical submissions.	Must	Direct requirement for the e-learning workflow.
As an administrator I want activity usage dashboards so I can monitor platform use.	Should	Useful for oversight, but not required immediately for core functionality.
As a student I want email notifications for deadlines/announcements so I stay informed.	Could	Adds convenience and improves user experience, but not critical.
As a student I want the system to load pages within 2 seconds so that the interface feels responsive.	Must	Performance (non-functional) is critical for usability.
As an instructor I want course certification issuance so completed students get credentials.	Should	Valuable feature, but can be implemented later.
As a guest user I want to see sample preview lessons so I can evaluate the platform before enrolling.	Could	Nice to have for marketing / conversion, but not essential initially.
As a system analyst I want data export to CSV so reports can be used offline.	Could	Useful for admin / analysts, but not blocking the MVP.
As a user I want a mobile app interface so I can access on the go.	Won't	Out of scope for initial release; defer to later version.

Assignment 3 – Timebox Plan

Timebox #	Duration	Key Objective(s)	Detailed Phases & Activities*	Deliverables / Outputs
Timebox 1	2 weeks	Establish core foundation and deliver first must-have features	**Kick-off (0.5 day):** agree timebox objectives, participant roles, dependencies **Investigation (~10–20%, ~2 days):** clarify user stories, acceptance criteria, risk review **Refinement (~60–70%, ~7–8 days):** design, build, initial testing **Consolidation (~10–20%, ~2 days):** final fixes, integration, quality checks **Close-Out (half day):** review what succeeded/not, carryover decisions, mini retrospective	Working features: browse courses, enroll in course, basic content access; acceptance criteria met
Timebox 2	2 weeks	Add assignments + performance / usability features	Kick-off, Investigation (2 days) to detail "submit assignments," "fast load time" Refinement (7–8 days): build, test, integrate Consolidation (2 days): polish, fix bugs Close-Out: review and retrospective	Feature increment: assignment submission module, performance optimisations
Timebox 3	2 weeks	Add instructor features and enhancements	Kick-off, Investigation (2 days): clarify upload materials, quizzes Refinement (7–8 days): build, test, integrate Consolidation (2 days): wrap up Close-Out: review, retrospective	Instructor module: upload content, quiz engine, basic feedback
Timebox 4	2 weeks	Add optional / secondary features	Kick-off, Investigation (2 days): define notifications, dashboards Refinement (7–8 days): develop/deliver Consolidation (2 days): final adjustments Close-Out: review and retrospective	Optional features (notifications, dashboard) and overall integration

* The percentages (Investigation, Refinement, Consolidation) follow DSDM structured timebox guidelines: Investigation ~10–20%, Refinement ~60–80%, Consolidation ~10–20%. Agile Business C... +1

Assignment 4 – Monetization Plan

1. Monetization Strategies (2–3 options)

Strategy	Description	Pros / Strengths	Risks / Challenges
Subscription / Membership Model	Users pay a recurring fee (monthly / yearly) to access a catalog of courses or premium features.	Predictable revenue stream; better retention; easier upselling	Churn risk; need to continuously deliver new value; pricing must be compelling
Freemium + Premium Features	Basic content or limited access is free; advanced features, courses, certifications, or ad-free experience are paid.	Low barrier to entry; encourages trial; upsell potential	Free users may never convert; balancing free vs premium features is delicate
Pay-Per-Course / Single Purchase + Bundling	Users pay a one-time fee to access a specific course or buy course bundles.	Simple to understand; no long-term commitment	Revenue is lumpy; difficult to predict; lower lifetime value unless you upsell
Commission / Revenue Sharing (Marketplace Model)	Platform hosts courses by third-party instructors and takes a commission or share of sales.	Scalable; low content creation burden; aligns incentives with instructors	Need enough instructors & traffic; margin split must be attractive; quality control required
Corporate / B2B Licensing & Training Partnerships	Sell licenses or access to enterprises, schools, or organizations for their employees or students.	Large contracts; stable revenue; higher willingness to pay	Long sales cycles; customization demands; more support overhead
Advertising & Sponsorships / Ad-Free Upgrade	Display ads in free content or partner with sponsors; offer a paid ad-free premium tier.	Monetizes free users; extra revenue stream	Risk of poor user experience; low ad revenue unless traffic is high
Certification / Credential Fees	Charge for official certificates, accreditation, or graded assessments.	Adds perceived premium value; can command higher margin	Must assure credibility; users may bypass certificate option
Usage / Credit System	Sell "learning credits" or tokens which users spend to unlock content or features.	Flexibility in pricing & bundling; scalable	Complexity in managing credit balances; risk users not using credits

> Many e-learning platforms combine multiple strategies (e.g. freemium + subscription + certificate fees) to diversify revenue. LearnWorlds +3

2. Financial Model (Example)

Here's a simplified financial model to illustrate how revenue, c
according to your region, market, and assumptions.)

Metric	Value / Assumption
Monthly subscription fee	RM 30 / user
Number of paying users (Year 1)	300 users
Monthly revenue from subscription	300 × RM 30 = RM 9,000
Annual subscription revenue	RM 108,000
One-time course sales	Assume 100 courses sold at
Certificate / credential fees	100 users × RM 20 = RM 2,(
Ad / sponsorship revenue	RM 1,200 / year
Total projected annual revenue	≈ RM 121,200
Cost items	Hosting and infrastructure (Content development / lice Marketing & operations (RM Staffing / support (RM 18,0(
Total costs	RM 65,000
Estimated profit / margin	RM 56,200 (≈ 46%)
Break-even point	~ 1–2 months of full subscri costs



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